Certificate-driven polytope-hull contraction for interval branch and bound

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Abstract

Interval branch-and-bound solvers compute rigorous enclosures of the solutions of nonlinear systems and of global optima. Their polytope-hull contractor bounds each of the n variables of a node over a linear relaxation by solving $2n$ small linear programs (LPs) with a floating-point LP solver, whose dual solutions are certified afterwards by the Neumaier–Shcherbina bound. That bound is valid for any vector of multipliers, so it makes any LP solver safe to use; yet a general-purpose solver still spends, inside each call, the work of making its own answer correct. We present a self-certifying simplex (SC) that omits that work, which the certificate makes unnecessary: it performs no verification of its own, does not refactorize before declaring an optimum, needs no phase 1, and solves the $2n$ LPs on one shared basis. A ladder of configurations in the open-source solver Ibex, from SoPlex as Ibex uses it, through the merging of twin rows, a tuned SoPlex and its simplex driven directly, to SC, attributes the gain to each step. On a bank of recorded passes, with the same input boxes and relaxations and the same contraction, a pass of SC on relaxations of up to 30 rows costs 0.39–0.43 times that of SoPlex as Ibex uses it, and 20–25% less than the simplex of SoPlex driven directly. Inside the search, the optimizer takes about 20% less CPU time than with SoPlex as Ibex uses it. On larger relaxations SC is slower than the simplex of SoPlex, and a hybrid that chooses the engine by the size of the relaxation keeps the best of both.

1 Introduction

Interval branch-and-bound solvers compute, with mathematical certainty, an enclosure of all solutions of a system of nonlinear equations and inequalities, or the global optimum of a constrained problem [22]. Their efficiency depends on *contractors*, operators that shrink the current box without losing solutions [7]. Among them, contractors based on linear relaxations are among the most effective: the constraints are linearized over the box, by convex interval Taylor forms [3] or affine arithmetic [25, 2], and each variable is minimized and maximized over the resulting polytope. In Ibex this is the polytope-hull contractor, the linear part of X-Newton, which we introduced in [3]; we later combined the affine and Taylor-based relaxations [2].

Each call solves up to $2n$ linear programs, and the rigor of the result cannot come from the LP solver, which works in floating point. Current implementations therefore treat the LP solver as a black box and certify its output a posteriori: the dual solution is turned into a safe bound by the Neumaier–Shcherbina procedure [23], and an infeasibility claim is accepted only if the returned

Farkas ray passes the same verification. This design has a cost that is easy to miss: the $2n$ LPs of a node are small and closely related, typically two or three pivots apart, and each is a separate call to a general-purpose solver, which sets up its vectors, refactorizes the basis before declaring an optimum and verifies its solution; on these LPs that work is a large part of the cost of a call (Section 5).

The certificate already makes any LP solver safe to use: with the relaxation written in a homogeneous form, with one interval-bounded slack per row, *any* multiplier vector yields a valid bound on the box and on every sub-box, whatever solver produced it. But a black box cannot take advantage of this, because the work it does to make its answer correct is internal to each call. We design the simplex around the certificate instead: since its multipliers never need to be correct, it performs no verification of its own and does not refactorize before declaring an optimum. Since every basis can also be made dual feasible by relocating its nonbasic variables, each of the $2n$ bound problems, and each problem after the box contracts, starts without a phase 1 (the search for a feasible starting basis), and the $2n$ problems share one basis. We call this simplex *self-certifying* (SC). How much it is worth depends on the size of the relaxation and on what else the contractor does, and measuring that is part of the contribution.

Positioning. The certificate itself is not new: that any multiplier vector yields a valid bound once the residual is evaluated in interval arithmetic is the principle of Neumaier and Shcherbina [23]. Domes and Neumaier [10] applied it to filtering with linear relaxations; in previous work [26] we wrote the relaxation in the homogeneous form used here and showed that the dual solutions of the bound LPs give the optimal preconditioner for an interval Gauss–Seidel projection, and Gleixner et al. [13] reuse the equivalent inequalities in sub-boxes. In all these approaches the multipliers are a by-product of an external solver and certification is a post-processing step, whose cost leads to applying it to a few promising variables [10] or only periodically [26]. Yamamura and coworkers [29, 30] observed that the optimal tableau of the parent remains dual feasible after a bisection, and resume an LP emptiness test by the dual simplex method; SC extends this warm start to the $2n$ bound problems, with rigorous certification. The heuristic of skipping bounds already attained by a primal point, used by Ibex, comes from Baharev, Achterberg and Rév [4].

Contributions. The paper makes three contributions.

- *A simplex designed for the certificate.* SC omits the verification, the refactorization before an optimum and the phase 1 of a general-purpose solver, and solves the $2n$ problems of a node on one shared basis (Section 4). The properties it relies on, validity of the certificate for any multipliers, exactness at an optimum and dual feasibility for free, follow from the Neumaier–Shcherbina bound and LP duality; we state them in Section 3.1 because the design rests on them, not as new results.
- *An attribution of the gain.* A ladder of configurations inside the same contractor of Ibex, from SoPlex as Ibex uses it, through the merging of twin rows (a simple preprocessing that halves the relaxation of square systems for every engine), a tuned SoPlex and its simplex driven directly, to SC and a hybrid, measures what each step contributes, separately on optimization problems and systems of equations and on small and large relaxations: on a bank of recorded passes, where every rung starts from the same boxes and relaxations, and inside the search, with one instance per family (Section 5).
- *The boundary of the approach.* SC pays on small relaxations, where the fixed cost of a call dominates (per pass, up to about 30 rows); beyond, it needs more pivots than SoPlex, and

each costs more. We explain why, and give a hybrid contractor that chooses the engine per node by the size of the relaxation (Sections 5 and 6).

2 Background and related work

Interval solvers and linear relaxations. Interval branch and bound alternates bisection with contraction. Ibex, on which we build, composes constraint propagation (HC4 [6]), shaving-based consistencies such as ACID [24], the interval Newton operator for square systems [21], and contractors based on linear relaxations [7], whose relaxation comes from convex interval Taylor forms [3], from affine arithmetic [25], or from both [2]. QUAD [19] and GloptLab [10] use linear relaxations in the same way.

Rigorous bounds for linear programs. A floating-point LP solver returns approximate solutions, and two lines of work make its results rigorous. The first certifies them a posteriori: any approximate dual vector yields a safe bound once the residual is evaluated in interval arithmetic, and an approximate Farkas ray proves infeasibility under the same verification [23, 16, 17]. The second solves the LP exactly, in rational arithmetic [1], or to arbitrary precision by iterative refinement [14], as in exact mixed-integer programming [8]. Both treat the LP solver as given; we keep the a posteriori certificate and change the solver. Minimizing and maximizing each variable over a relaxation is also the optimization-based bound tightening of global optimization solvers, where rigor is not required [27, 5, 13].

Simplex techniques. SC uses standard ingredients, described by Koberstein [18] for the dual method: the bounded-variable dual simplex with Harris' ratio test [15], and the revised method with a sparse LU factorization updated by the method of Forrest and Tomlin [12], as in SoPlex [28]. What is specific to our setting is their use for $2n$ closely related LPs over the same constraints, with a certificate attached to every basis.

3 The contractor

This section describes the contractor independently of the engine that solves its LPs: the relaxation and its certificate (Section 3.1), the pass over the bounds (Section 3.2) and the merging of twin rows (Section 3.3). Section 4 then describes SC, and Section 5 compares it with external LP solvers inside this same contractor. Only SC is new; the rest is the common ground on which the engines are compared.

3.1 The homogeneous relaxation and its certificate

Let $\mathbf{x} \in \mathbb{IR}^n$ be the current box. A linearizer returns a matrix $A \in \mathbb{R}^{m\times n}$ of floating-point numbers and an interval vector $\mathbf{b} \in \mathbb{IR}^m$, possibly with infinite bounds, such that

$$x \in \mathbf{x} \text{ is a solution } \implies Ax \in \mathbf{b}. \tag{1}$$

We introduce one slack per row, $s = Ax$, and write the relaxation in the homogeneous form

$$\bar{A}z = 0, \qquad z = (x, s) \in \mathbf{z} := \mathbf{x} \times \mathbf{s}, \qquad \bar{A} := [\, A \;\; -I_m \,], \tag{2}$$

where $\mathbf{s}_i := \mathbf{b}_i \cap \square(A_{i:}\,\mathbf{x})$ and $\square(\cdot)$ denotes evaluation in interval arithmetic. We call $P(\mathbf{x}) := \{z \in \mathbf{z} : \bar{A}z = 0\}$ the *relaxation polytope*.

Lemma 1. *If $x^* \in \mathbf{x}$ is a solution, then $(x^*, Ax^*) \in P(\mathbf{x})$.*

Proof. $A_{i:}x^* \in \mathbf{b}_i$ by (1) and $A_{i:}x^* \in \square(A_{i:}\mathbf{x})$ by the inclusion property of interval arithmetic; the equation holds by construction of s. □

The certificate combines the m rows of $\bar{A}z = 0$ linearly, with one *multiplier* per row, into a single equation $\gamma^T z = 0$ that every point of the polytope satisfies. Any vector of multipliers $\lambda \in \mathbb{R}^m$ gives a valid equation; the best ones for bounding z_k are the dual solutions of the LP that minimizes or maximizes z_k over the polytope, and the simplex of Section 4 produces them.

Definition 2 (Certificate)**.** For a vector of multipliers $\lambda \in \mathbb{R}^m$, let $\gamma \in \mathbb{IR}^{n+m}$ be an interval vector that contains $\gamma := \bar{A}^T\lambda$, computed in outward-rounded interval arithmetic. For $k \in \{1, \ldots, n+m\}$ with $0 \notin \gamma_k$, the *certified enclosure* of z_k is

$$\mathbf{C}_k(\lambda, \mathbf{z}) := -\Big(\textstyle\sum_{j\neq k} \gamma_j\, \mathbf{z}_j\Big) \Big/ \gamma_k,$$

and the *residual* is $\mathbf{R}(\lambda, \mathbf{z}) := \sum_j \gamma_j\, \mathbf{z}_j$.

The idea is a single line. Every point z of the polytope satisfies $\bar{A}z = 0$, hence $\gamma^T z = \lambda^T \bar{A} z = 0$ for any λ, that is, $\gamma_k z_k = -\sum_{j\neq k} \gamma_j z_j$. Bounding the right-hand side over the box in interval arithmetic, and dividing by γ_k, encloses z_k; and if the residual $\mathbf{R}$ excludes zero, no point of the box can satisfy $\gamma^T z = 0$, so the polytope is empty. A contraction step therefore replaces $\mathbf{z}_k$ by $\mathbf{z}_k \cap \mathbf{C}_k(\lambda, \mathbf{z})$, and an emptiness test checks $0 \notin \mathbf{R}(\lambda, \mathbf{z})$. All statements below hold for the floating-point A returned by the linearizer; rounding in γ and $\mathbf{C}_k$ is covered by outward rounding.

Proposition 3 (Validity)**.** *Let $\lambda \in \mathbb{R}^m$ be arbitrary. For every $z \in P(\mathbf{x})$ and every k with $0 \notin \gamma_k$, $z_k \in \mathbf{C}_k(\lambda, \mathbf{z})$; if $0 \notin \mathbf{R}(\lambda, \mathbf{z})$, then $P(\mathbf{x}) = \emptyset$ and $\mathbf{x}$ contains no solution. Both claims hold on every sub-box $\mathbf{x}' \subseteq \mathbf{x}$ with the matrix $\bar{A}$ computed on $\mathbf{x}$.*

Proof. For $z \in P(\mathbf{x})$ and the exact $\gamma = \bar{A}^T\lambda \in \gamma$, $\gamma^T z = \lambda^T \bar{A} z = 0$; both claims follow from the inclusion property of interval arithmetic and Lemma 1. The linearization on $\mathbf{x}$ remains valid on $\mathbf{x}'$, so the polytope over $\mathbf{x}'$ lies in $P(\mathbf{x})$. □

No property of λ is used: it need not be optimal, feasible or accurate, and errors in it can only weaken the enclosure. Written back in the variables x, $\inf \mathbf{C}_k(\lambda, \mathbf{z})$ is the Neumaier–Shcherbina safe lower bound of $\min x_k$ subject to $Ax \in \mathbf{b}$, $x \in \mathbf{x}$, with approximate dual λ [23], and the row-wise solution of the preconditioned system of Domes and Neumaier [10, §5]; validity on sub-boxes is used by Gleixner et al. [13] and in our earlier work [26]. The homogeneous form, introduced in [26], adds one detail: each slack is bounded by $\square(A_{i:}\mathbf{x})$ even when a side of $\mathbf{b}_i$ is infinite, so a multiplier of the "wrong" sign on a one-sided row gives a weaker but finite bound, and literally any λ certifies. Two standard facts complete what the design uses. At an optimal basis of LP_k with z_k basic, $\inf \mathbf{C}_k(\lambda^*, \mathbf{z})$ equals the optimal value in exact arithmetic, by LP duality [26, Prop. 1]; in floating point the excess is at most $\sum_j \mathrm{wid}(\gamma_j)\,\mathrm{mag}(\mathbf{z}_j)$ plus the effect of the inaccuracy of λ, as for the Neumaier–Shcherbina bound. And every basis can be made dual feasible for free [18]:

Proposition 4 (Dual-feasible start)**.** *The slack basis $B = -I_m$ is dual feasible for the bound problem of every structural variable z_k, $k \le n$, with $\inf \mathbf{z}_k$ finite. If every $\mathbf{z}_j$ is bounded, any basis becomes dual feasible after placing each nonbasic z_j at its lower bound if its reduced cost $d_j > 0$ and at its upper bound if $d_j < 0$.*

Proof. With the slack basis the cost of every basic variable is zero, so $\lambda = 0$ and $d = e_k$, and z_k at its lower bound has the right sign. In general d does not depend on where the nonbasic variables sit, and the relocation enforces the sign conditions whenever the required bound is finite. □

When the objective changes from one bound to the next, or the box contracts, relocation thus restores dual feasibility without a phase 1, which is what lets one basis serve the $2n$ problems. Yamamura and coworkers [29, 30] used the case of a box change to resume an LP emptiness test after a bisection. Section 4 explains how the simplex produces λ.

Why not a rigorous simplex. Since any multipliers give a valid bound, rigor is needed only in the final evaluation of γ and $\mathbf{C}_k$, not in the pivoting. Pivoting in interval arithmetic pays the width growth of interval Gaussian elimination [21], and solving each LP exactly or to high precision [1, 14] buys an exactness that the bound does not need; an early version of our contractor that pivoted in interval arithmetic was 2.8 times slower, with the same contraction.

3.2 One pass over the bounds

At each node the relaxation is computed, written in the form (2), and loaded once. The contractor then makes one pass over the up to $2n$ bound problems

$$\mathrm{LP}_k: \quad \min\ z_k \quad \text{s.t.} \quad \bar{A}z = 0,\ z \in \mathbf{z}, \tag{3}$$

and their maximizations, obtained with $-z_k$, each LP warm-started from the final basis of the previous one. When an LP ends, its bound is certified once, from the multipliers of the final basis, and the box is contracted before the next LP starts. Two rules decide which LPs are solved and in which order. A bound is *skipped* when a primal point obtained earlier in the pass already attains it, since its LP would return that value; this rule comes from the polytope-hull contractor of Ibex [4]. Among the remaining bounds, we solve first, as the contractor of Ibex does, the one nearest to the last primal point, whose optimum lies closest to the vertex where the previous LP ended, so that the next LP needs fewer pivots.

If the residual $\mathbf{R}(\lambda, \mathbf{z})$ of an LP excludes zero, the node is empty and the pass stops. No other outcome stops it. An LP that ends without an optimum, at its iteration limit, still has the multipliers of the basis where it stopped, which certify a weaker but valid bound (Proposition 3); an LP that reports an infeasibility whose ray fails the residual test leaves its bound uncontracted, and is solved once more from a cold start. The polytope-hull contractor of Ibex 2.8.9 instead abandons the pass in both cases; we corrected it, the correction has been submitted to the Ibex distribution, and every engine of Section 5 uses it, including the reference. Ibex limits SoPlex to 100 iterations per LP; from the tuned SoPlex on (Section 5), every engine, SC included, uses $20m + 200$, where m is the number of rows.

3.3 Merging twin rows

Rows of the relaxation that are proportional, within a relative tolerance of 10^{-9}, are merged into one row whose slack interval is the intersection of their scaled intervals; the polytope does not change. The affine linearizer writes each equation as two one-sided rows with the same normal, so merging halves m on square systems, and the combined relaxation of Section 5 writes some rows twice, with opposite signs, far apart in the matrix. Merging is done once per node, before any LP, and gives a smaller LP to SC and to an external solver alike.

4 A simplex for the $2n$ bound problems

For each structural variable x_k the pass of Section 3.2 needs the two bounds of LP_k (3). A basis is a set of m columns of $\bar{A}$ forming a nonsingular matrix B; with cost vector $c = \pm e_k$, the multipliers are $\lambda^T = c_B^T B^{-1}$ and the reduced costs $d = c - \bar{A}^T\lambda$. For the bounded-variable simplex, a basis with each nonbasic variable at one of its bounds is *dual feasible* if $d_j \geq 0$ for nonbasic variables at their lower bound and $d_j \leq 0$ for those at their upper bound, and *primal feasible* if the basic variables lie within their bounds. This section describes how the simplex solves the $2n$ problems of a node; the design choices are evaluated in Section 5.

What the certificate makes unnecessary. Since the certificate of Section 3.1 is valid for any λ (Proposition 3), the simplex never has to return a correct solution, only a good basis. A general-purpose LP solver, before declaring an optimum, refactorizes the basis to remove the error accumulated by the updates of its factorization, recomputes its vectors and verifies the solution, and the contractor then verifies it once more through the certificate. SC uses the final basis as it is: an inaccurate λ yields a weaker bound, never an invalid one, so the error of the factorization is controlled only as far as pivoting requires. Likewise, since every basis can be made dual feasible (Proposition 4), no phase 1 is needed when the objective changes or the box contracts, and the $2n$ problems share one basis. Section 5.1 measures what these savings are worth.

4.1 Starting and warm-starting

The first LP of a pass starts from the slack basis $B = -I_m$, which is dual feasible for every LP_k (Proposition 4), and is solved by the dual method, with no phase 1. Between two consecutive LPs the objective changes and the contracted bound changes the box, and two warm starts are possible from the previous optimal basis. The *dual* one relocates every nonbasic variable to the bound indicated by the sign of its new reduced cost, which restores dual feasibility, and continues with the dual method; the *primal* one keeps the nonbasic variables where they are and reoptimizes with primal steps, as an LP solver does. Under the dual method the certificate is available at every iteration, and it never gets worse:

Proposition 5 (Monotone certified bound)**.** *Run the dual simplex method on* LP_k *from a dual-feasible basis. At every iteration* t *in which* z_k *is basic,* $\inf \mathbf{C}_k(\lambda_t, \mathbf{z})$ *is a valid lower bound on* z_k *over* $P(\mathbf{x})$ *and, in exact arithmetic, it is nondecreasing in* t*.*

Proof. Validity is Proposition 3. With z_k basic, $d = e_k - \bar{A}^T\lambda_t$ has $d_k = 0$, so $\gamma_k = 1$, $\gamma_j = -d_j$ for $j \neq k$, and $\inf \mathbf{C}_k = \sum_{j\neq k} \min_{z_j \in \mathbf{z}_j} d_j z_j = \min_{z\in\mathbf{z}} d^T z$, the Lagrangian dual function of LP_k at λ_t, which the iterates of the dual simplex method do not decrease. □

Both warm starts reach an optimal basis and give the same bound; they differ in cost. A nonbasic variable that must move to its opposite bound costs the primal warm start an iteration, while relocation moves it for free but forces the basic values to be recomputed once per LP. On systems of equations every row is a narrow slab (its slack interval covers 13% of the range of the row over the box, on average), and a median of 1.9 nonbasic variables per LP change side, against 0.3 on optimization problems, whose slack intervals cover 48% of the range. SC therefore uses the primal warm start in the optimizer, where the dual one is 4% slower, and the dual one on systems of equations, where both cost about the same; with `compo`, where neither wins on every system, it alternates the two on the first nodes and keeps the one with the lower measured CPU time per LP, trying the other again every 25 nodes (so the choice, unlike the rest of SC, depends on timing).

4.2 Simplex iterations

We follow the bounded-variable simplex of [18]. Each iteration exchanges one basic and one nonbasic variable using the pivot column $\alpha_q = B^{-1}\bar{A}_{:q}$ of the entering variable (FTRAN), the pivot row $\alpha_r^T = \rho_r^T \bar{A}$ with $B^T\rho_r = e_r$ (BTRAN), and updates $x_B \leftarrow x_B - \theta_p\alpha_q$, $\lambda \leftarrow \lambda + \theta_d\rho_r$ and $d \leftarrow d - \theta_d\alpha_r$, so that λ, which the certificate needs, is always at hand.

Dual steps. In a dual iteration the basis is dual feasible and some basic variable violates its bounds. The leaving variable is the basic variable with the largest violation. The entering variable comes from the ratio test over the pivot row: among the nonbasic z_j whose reduced cost would change sign in the wrong direction, the one with the smallest $|d_j/\alpha_{rj}|$ keeps the basis dual feasible; among the candidates within a relative band of that ratio we take the largest $|\alpha_{rj}|$ (Harris' rule), which avoids tiny pivots.

Rules that did not pay. We also implemented the standard refinements of the dual method [18]: devex pricing [15] for the leaving variable, the long-step (bound flipping) ratio test [20], and a limit of 60 pivots per LP, which the certificate makes safe (Proposition 5). Each helps on some families and hurts on others: together they make the Bratu systems 1.3–1.5 times faster and the BroydenBanded ones up to 2.6 times slower, and on average they leave the time of the search unchanged (Section 5). SC therefore uses the textbook rules above. On the small LPs of a pass, the choice of pivots matters less than the work that surrounds them.

Primal steps, tolerances. Primal steps enter the nonbasic variable with the largest reduced cost of the wrong sign (Dantzig's rule; with `compo`, whose primal warm starts are longer, devex weights); pivots below a stability threshold are filtered out before choosing the leaving variable, since without this filter a rejected pivot was indistinguishable from optimality. The feasibility tolerances are 10^{-9}, relative to the value tested, and 10^{-6} with `compo` on systems of equations, whose nearly parallel rows make the LPs degenerate. A tolerance is a matter of efficiency, not of rigor: too wide, it weakens the contraction; too tight, the warm-started simplex reports infeasibility on boxes that are not empty, the ray fails the residual test, and SC solves that LP again from the slack basis.

4.3 The engine

SC keeps a sparse LU factorization of B, computed by the factorization code of SoPlex and updated after each pivot by the Forrest–Tomlin method [12], which reuses the FTRAN of the entering column. It refactorizes every 50 pivots and recomputes x_B every 32, to limit drift.

4.4 Certification

When an LP terminates, the certificate $\gamma = \bar{A}^T\lambda$ is computed once in outward-rounded interval arithmetic, over the rows with $\lambda_i \neq 0$, and $\mathbf{C}_k$ and $\mathbf{R}$ follow in $O(n+m)$. When a dual iteration finds a violated basic variable and no entering variable in its row r, $\lambda = \rho_r$ is a Farkas ray, and the residual test turns it into a proof of emptiness. Certifying a bound costs the same as the Neumaier–Shcherbina post-processing of an external solver's dual (1.3–3.1 μs per LP on three optimization problems); the difference between the approaches is not the cost of a certificate but where λ comes from.

5 Experimental evaluation

A ladder of configurations. All configurations run inside the *same* contractor: the same pass over the $2n$ bounds, with the same skipping rule, the same certificate $\gamma = \bar{A}^T\lambda$ applied to whatever multipliers the engine returns (its dual solution, or its Farkas ray if it reports infeasibility), and the same policy when an LP ends without an optimum: the multipliers where it stopped certify a weaker bound, and the pass continues (Section 3.2). The configurations differ in one step at a time:

- R0 *SoPlex as Ibex uses it*: SoPlex 4.0.2 through the LP interface of Ibex, which stores the n variable bounds as rows, with presolving on, the tolerance 10^{-9}, the nearest bound first as in the contractor of Ibex, and at most 100 iterations per LP;
- R1 + *merging twin rows* (Section 3.3);
- R2 + *a tuned SoPlex*: its own instance holding the box as column bounds, presolving off, the tolerance 10^{-6} for systems of equations and 10^{-9} for optimization, and the limit $20m + 200$;[1]
- R3 *the bare simplex*: the simplex of SoPlex driven directly through its `SPxSolver` class, the LP kept loaded, only the objective and the column bounds changed between LPs, nothing verified; it is the cheapest way to use SoPlex;
- R4 *SC* (Section 4);
- R5 *the hybrid*: SC on relaxations of at most 30 rows, the bare simplex beyond.

Each rung is compared with R0. HiGHS 1.15.1 and Gurobi 12.0.3, with one thread, the dual simplex, the basis kept between LPs and no presolve, cost 2.3–4.6 times as much per pass as the tuned SoPlex, so we leave them out of the ladder.

Setting. Contractor and solvers are implemented in Ibex 2.8.9, with the two relaxations of our previous work [2]: the affine relaxation [25], and the combined relaxation (`compo`), which adds to the affine rows those of X-Newton [3], one per constraint and corner. Optimization problems use the affine relaxation, since in preliminary experiments `compo` cost more than it saved with every engine; systems of equations are evaluated with both, since `compo` is the better one on large systems [2] and the affine one keeps most relaxations small. Per node, after merging, the affine relaxation has a median of 5 rows (quartiles 3–9) on optimization problems and 16 (9–30) on systems of equations; `compo` has 56 (36–94) on systems. For systems of equations we use the solver composition of Ibex (HC4, ACID, interval Newton when the system is square, and the linear contractor within a fixed point with HC4) with the LSmear bisector, on the 242 systems of the Ibex solver benchmark that some configuration solves within 3600 s, 2 seeds, 60 s per run; for optimization, IbexOpt on the 240 problems of the Ibex optimization benchmark that some configuration solves within 3600 s, 4 seeds, 60 s per run. All runs were made on two Intel Xeon E5-2620 v4 processors (2.1 GHz, 16 cores) with 64 GB of memory, under Linux 6.17, eight runs at a time with at most 6 GB each; the code is compiled with GCC 13.3 (`-O3`) with the GAOL 4.2.0 interval library. CPU times are process times. The code, the banks of passes and the scripts are public (see the statement of code and data availability).

[1] Each of these choices is the best one for SoPlex on these LPs: presolving each new relaxation costs more than it saves on LPs of a few pivots, 10^{-9} makes SoPlex report spurious infeasibility on relaxations with nearly equal rows, and 10^{-6} weakens its dual on optimization problems. Turning off scaling, or forcing the row or column representation or the primal algorithm, did not help, and SoPlex 7.1.6 was 1.2–1.9 times slower than 4.0.2 on a sample of eight problems.

Table 1: Time per pass of each rung over that of SoPlex as Ibex uses it (R0), on the bank of passes (1 000 recorded calls per instance, same input boxes for every rung), by program and by the mean number of rows of the merged relaxation: affine relaxation up to 30 rows, `compo` above. Quotient of the means of the times normalized per instance (see text). The six systems whose affine relaxation exceeds 30 rows are discussed in the text. On the systems R5 equals R4 or R3 by construction. Last rows: SC over R3 directly, with the same metric, and instances where SC is cheaper than R3.

	optimization	systems of equations	
rung	(74)	≤ 30 rows (30)	> 30 rows (41)
R1 + merging twin rows	0.93	0.84	0.85
R2 + tuned SoPlex	0.68	0.66	0.71
R3 bare simplex	0.52	0.52	**0.56**
R4 SC	**0.39**	**0.43**	0.83
R5 hybrid	0.40	0.43	0.56
R4 / R3	0.75	0.80	1.42
SC cheaper than R3	69 / 74	25 / 30	7 / 41

Rigor. SC is rigorous by Proposition 3; we also checked it against the LP hull. On four systems, for each of the 2 834 bounds where SC contracted more than the polytope-hull contractor of Ibex on the same box and polytope, we solved the bound LP over the input box with a fresh instance of SoPlex; SC attained this hull in 98% of these bounds and went beyond it in 19, all within the 10^{-9} feasibility tolerance of the oracle.

5.1 Cost of a pass

To measure the contractor without the noise of the search, we built a bank of passes: for 110 instances, 36 systems of equations (one per family) and 74 optimization problems (one per family and size class, among those on which R0 takes more than 1 s), we recorded the input boxes of 1 000 calls of the contractor spread along a search with SC, with the affine relaxation, and replayed every call with each rung from the same input box and relaxation. Because the affine relaxation of only six systems exceeds 30 rows, a second bank does the same with `compo`, the best relaxation on large systems, for 41 systems whose relaxation has more than 30 rows (31 to 687, median 63; one per family, R0 above 1 s). Three optimization problems were left out: on pentagon and ex8_2_4, SoPlex as Ibex uses it aborted with an internal exception, and on concon the affine linearization failed for every rung. Table 1 gives the time per pass of each rung relative to R0. For each instance we normalize both times by the larger one, $t(A)/\max(t(A), t(\mathrm{R0}))$ and $t(\mathrm{R0})/\max(t(A), t(\mathrm{R0}))$, average each over the instances, and report the quotient of the two means; the normalization keeps a few long instances from dominating.

On optimization problems and on systems of up to 30 rows, a pass of SC costs 0.39–0.43 times that of R0, and 25% and 20% less than the bare simplex (geometric means of the ratio 0.74 and 0.78; SC is cheaper on 69 of the 74 optimization problems and 25 of the 30 systems). Before SC, merging saves 7–16%, tuning SoPlex 21–27% of what remains, and driving its simplex directly 21–24% more. The median pass on optimization problems takes 260 μs with R0, 136 with the bare simplex and 88 with SC; on small systems, 192, 108 and 84 μs (medians weigh the many fast instances, where SC gains most).

The gain of SC lies in the cost of an LP, not in its pivots. The tuned SoPlex and the bare

simplex need a median of 1.8 pivots per LP, SC 2.8, because SoPlex often ends an LP at a vertex that is already optimal for the next bound (Section 6); yet an LP of SC costs 11.6 μs on optimization problems and 16.2 on small systems, against 16.0 and 21.7 for the bare simplex. What SC does not do is what makes the difference: the bare simplex keeps the LP loaded and verifies nothing, so the remaining gap is the refactorization that SoPlex makes before declaring an optimum and the setup of each solve. Re-inserting into SC, at every LP, the work that a call of the tuned SoPlex performs, on separate vectors so that the contraction does not change, brings a pass of SC from 0.55–0.60 to 0.89–0.93 times that of R2 (geometric means on 100 of the 104 instances; the emulation failed on four): refactorizing at every LP accounts for 48–55% of the gap (in logarithm), verifying the solution for 21–23%, and recomputing the vectors and copying the solution for 9–10%.

Merging pays more with SC. Ordered as a ladder, each step is measured on the engine of its rung, and merging is measured on R0, the most expensive one. On the same bank, merging saves 7% of a pass of R0 on optimization problems and 16% on systems of equations, but 21% and 43–47% of a pass of SC. A call of SoPlex carries a fixed cost that does not depend on the rows; SC has almost none, so its time follows the size of the LP, and halving the rows of a square system nearly halves it. Merging is a simple preprocessing, not a contribution of this paper, but it is worth most with an engine that has removed the fixed cost of a call.

Same contraction. All rungs empty the same boxes and contract alike: the sum of the widths of the box shrinks by 25.5% per call on average on optimization problems and by 18.6–18.7% on small systems for R1 to R4, and within 1% on all but 6–9 of the 110 instances. The exception is R0, whose limit of 100 iterations ends 8.5% of the LPs of small systems without an optimum (0.9% after merging; less than 0.1% with $20m+200$); the multipliers where it stops certify a weaker bound, and R0 by 18.2%. On Geneig-icse the difference is 0.23 against 0.34. On the `compo` bank the LPs are larger and more degenerate, and with the tolerance of 10^{-6} the certified bounds of the engines differ slightly more: SC contracts by 15.6%, the tuned SoPlex by 15.9% and the bare simplex by 16.1%, while the limit of R0 ends 5% of the LPs without an optimum.

Large relaxations. On the systems whose relaxation exceeds 30 rows the order of the last two rungs is reversed (Figure 1): on the `compo` bank a pass of SC costs 1.42 times that of the bare simplex (1.48 in geometric mean) and is cheaper on only 7 of the 41 systems, and on the six affine ones it costs 1.56 times and loses on all. Merging and tuning SoPlex keep their weight. There an LP needs a dozen pivots, the fixed cost of a call no longer dominates, and SC needs more pivots, each more expensive (medians on the `compo` bank: 1.16 times the pivots per LP of the bare simplex, at 1.25 times its cost per pivot; Section 6); the hybrid hands these relaxations to the bare simplex. Devex pricing, the long-step ratio test and the limit of 60 pivots (Section 4.2) bring a pass of SC from 0.75 to 0.55 times that of R0 on the six affine systems (the bare simplex: 0.47), but leave it at 1.46 times the bare simplex on the `compo` bank (geometric mean), and change little on the smaller relaxations.

5.2 The whole solver

Table 2 runs the same ladder inside the search. To weight families equally, we keep one instance per family and size class, the one with the largest relaxation; families are the prefix of the name on systems of equations, while on optimization problems every problem is its own family, except for reformulations of the same problem (`bis`, `-1`). To populate both size classes of the systems, we

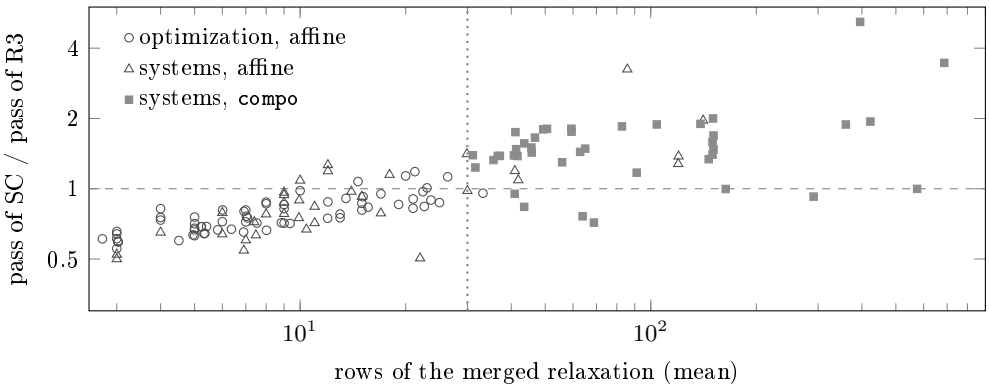


Figure 1: Time per pass of SC over that of the bare simplex (R3) on the two banks, one point per instance, against the size of the relaxation. Below the dashed line SC is cheaper; the dotted line is the threshold of the hybrid.

Table 2: CPU time of each rung over that of SoPlex as Ibex uses it (R0), inside the search, one instance per family and size class, for runs where one of the two configurations takes more than 1 s (resp. 5 s); number of families in parentheses. On the classes up to 30 rows the hybrid is not reported: it runs SC on almost every node. Systems of equations pool the affine and `compo` relaxations. The search trees of R1 to R5 agree within 3%.

	optimization		systems of equations			
	≤ 30 rows (189)		≤ 30 rows (65)		> 30 rows (52)	
rung	> 1 s	> 5 s	> 1 s	> 5 s	> 1 s	> 5 s
R1 + merging twin rows	0.97	0.95	0.96	0.95	0.91	0.89
R2 + tuned SoPlex	0.90	0.90	0.89	0.86	0.81	0.77
R3 bare simplex	0.83	0.84	0.84	0.83	**0.78**	**0.75**
R4 SC	**0.78**	**0.80**	**0.83**	**0.82**	0.93	0.88
R5 hybrid	–	–	–	–	**0.78**	**0.74**

pool their runs with the affine and `compo` relaxations, each (instance, relaxation) classified by its own size. Times are normalized per run as on the bank; unfinished runs count as 60 s.

The ladder keeps its shape, but the gains are diluted: the pass is only part of the cost of a node, with the linearization, HC4, ACID and interval Newton. On optimization problems the solver takes 20–22% less CPU time with SC than with R0, and 5–6% less than with the bare simplex. On small systems SC and the bare simplex tie (0.83 against 0.84), because the pass weighs less in the system solver. Counting the calls of the contractor in a search with SC and multiplying them by the cost of a pass on the bank, the pass takes a median of 16% of the CPU on optimization problems (interquartile range 7–26%) but 4.5% on systems of equations (2–13%), where the linearization, HC4, ACID and interval Newton dominate a node; saving 20% of 4.5% is about 1%. On large systems SC is 17–19% slower than the bare simplex, and the hybrid, which runs the bare simplex there, matches it. Only three optimization families exceed 30 rows (33–49 rows); SC remains the best on them (0.82 against 0.97 for the bare simplex), so the hybrid could keep SC on optimization problems at any size. Devex, the long-step ratio test and the limit of 60 pivots change these ratios by at most 0.01.

The threshold of the hybrid. From the runs of every rung we evaluated offline the hybrid that takes, for each (instance, relaxation), SC if its mean relaxation has at most T rows and the bare simplex otherwise. Against the bare simplex, SC gains below 15 rows (0.86–0.92 on optimization problems, 0.95–0.98 on systems), ties from 15 to 30, and loses from 40 rows on (1.06 at 40–50 rows, 1.25 beyond 120 on systems). The total time of the hybrid is flat for T between 15 and 30, and we use $T = 30$; the contractor applies it per node, with the size of the relaxation of the previous node. Run inside the search on the systems of equations with `compo` (two seeds), the hybrid with $T = 15$, 20 and 30 costs 0.98–1.01 times the bare simplex on the small systems and 1.00 on the large ones, where SC alone costs 1.19: the choice per node loses nothing, and the threshold does not matter inside this range.

6 Discussion

Why SC wins. SC does not contract with a better polytope: every rung solves the same relaxation and feeds the same certificate, and from R1 on they contract alike. Its advantage is the cost of an LP. The ladder separates two parts of the cost of a call of SoPlex as Ibex makes it: the interface and its options (R1 to R3), which a careful integration of SoPlex removes, and what the simplex of SoPlex itself does at every LP, which only a simplex designed for the certificate can omit. SC pivots more than SoPlex, but it neither refactorizes before declaring an optimum nor verifies its answer, needs no phase 1 when the objective or the box changes, and keeps one basis for the whole pass. On small relaxations, where an LP needs two or three pivots, that saves 20–25% of a pass of the bare simplex.

Where it loses. The saving is a fixed cost per LP, so its weight falls as the pivots per LP grow, and beyond 30 rows they dominate. There SC needs more pivots than SoPlex, for two reasons that multiply. On Katsurabis-14 with `compo`, a pass of SC costs 1.6 times that of the tuned SoPlex on the bank, and the gap lies almost entirely in the pivots: 26.7 against 17.5 per LP, at 14.9 against 14.1 μs per pivot. The starting point, $\times 1.3$: started from the final basis of SoPlex instead of its own, SC drops from 27 to 21 pivots per LP and solves without a pivot every LP that SoPlex solves without one (73% from its own basis), because on a large degenerate polytope the optimum of a bound is a face, and SoPlex stops inside it at vertices that serve the next bound better. The pivoting rules, $\times 1.15$: 20.8 against 18.1 pivots from the same basis. The first reason is the most specific to this setting and the one we could not fix: on optimization problems SoPlex ends 13–40% of its LPs at a vertex that is already optimal for the next bound, and started from those bases SC needs as few pivots, but neither a tie-break toward the corners of the box inside the optimal face nor a short walk inside that face (up to ten primal steps that keep the optimum) made SC end at such vertices. The second is engineering, and the pricing of SoPlex would remove it; the cost per pivot, $\times 1.4$ in an earlier version of SC, is now within 6%. Until then the hybrid uses each engine where it is better; on systems of equations, whose best relaxation is `compo`, this is what makes the redesigned contractor pay.

What did not pay. Several ideas that use the certificate or the tableau beyond the pass did not pay on the whole benchmark. Certificates of an ancestor box remain valid (Proposition 3) but rarely prune what the node does not prune anyway. Truncating each LP when its anytime bound (Proposition 5) is close to what the LP can still gain saves pivots ($\times 0.84$) but weakens the contraction (nodes $\times 1.39$–1.60). Propagating with HC4 in the middle of a pass, cuts at the LP optima, and a Lagrangian bound improved by monotonicity reduce the nodes or tighten bounds,

but cost more than they save. Inheriting the basis of the parent node cuts the pivots of the first LP of a node by a factor of three or more, but the first LP is a small part of the node. Earlier versions of SC kept the whole tableau explicitly and perturbed the zero reduced costs of the highly dual-degenerate bound problems [18]; with the other components in place both made SC slower. Doing more work inside a node did not change the search tree; what changed it was what is computed: the merged relaxation, the stability filter of the ratio test, the Farkas ray of the infeasible row, the tolerance.

Limits. SC removes the external LP solver from the contractor, not from the optimizer: the upper-bounding of IbexOpt still solves its LPs with SoPlex. On problems with very narrow equalities the upper-bounding becomes the bottleneck: on bearing, whose eight equalities are relaxed to $|h| \leq 10^{-8}$, feasible points are found only in boxes of width 10^{-4} to 10^{-3}, and the search tree changes by two orders of magnitude between seeds; we excluded it, with alkyl and alkylbis, from the means. Relaxations of several hundred rows are still expensive for every engine: the systems of 400 to 1 000 variables of the benchmark exceed the limits of 60 s and 6 GB with all of them. And the gain of SC on the whole system solver is small, because the pass is a small part of its nodes (4.5% of the CPU in the median). One use of the certificate is left open: when the warm-started simplex declares an infeasibility that the ray does not certify, SC solves the LP again from the slack basis (Section 4.2), but the multipliers of the basis where it stopped, moved along the ray as long as the bound improves, already give a certified bound without that second LP.

7 Conclusion

A polytope-hull contractor solves $2n$ small, closely related LPs per node and certifies each bound with the Neumaier–Shcherbina bound, which is valid for any multipliers. A general-purpose LP solver cannot take advantage of this, because the work it does to make its answer correct is internal to each call. A simplex designed around the certificate omits that work: it verifies nothing, does not refactorize before declaring an optimum, needs no phase 1 and solves the $2n$ problems on one shared basis. Measured as a ladder inside the same contractor, from the same boxes and relaxations and with the same contraction, a pass of this simplex costs 0.39–0.43 times that of SoPlex as Ibex uses it on relaxations of up to 30 rows, and 20–25% less than the simplex of SoPlex driven directly. Inside the search, the optimizer takes 20–22% less CPU time than with SoPlex as Ibex uses it, and 5–6% less than with the bare simplex. On larger relaxations the pivots dominate and SoPlex needs fewer, mostly because the vertices where it ends an LP are better starting points for the next bound; a hybrid that chooses the engine by the size of the relaxation keeps the best of both. How to choose, inside the optimal face of a degenerate polytope, the vertex that serves the next bound is the open question.

Code and data availability

The code and data used in the experiments are archived at Zenodo (**[TODO:** DOI]); they are the branch `dfb-sc-paper` of `https://github.com/rilianx/ibex-lib`. SC and the contractor are in `examples/dfb`; `results/escalera_2026-11-06` holds the banks of passes (input boxes and relaxations), the raw results of every run, the scripts that ran them and those that produce the tables, with instructions to rebuild the binaries. The fix to the polytope-hull contractor of Ibex used in the baseline was submitted upstream (`https://github.com/ibex-team/ibex-lib/pull/579`).

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